For example, if $R = \{(1, a), (2, b), (2, c), (3, a)\}$, then

$$R^{-1} = \{(a, 1), (b, 2), (c, 2), (a, 3)\}.$$

It is easy to see that the domain of R becomes the range of R^{-1} and the range of R becomes the domain of R^{-1}.



An equivalence relation on a set decomposes it into a disjoint union of its subsets (equivalence classes). Such a decomposition is called a partition. This is explained in the following example.

For $a, b \in \mathbb{Z}$, aRb if and only if $a - b = 3k$, $k \in \mathbb{Z}$ is an equivalence relation on $\mathbb{Z}$.

$$\begin{aligned} Z_0 &= \{x \in \mathbb{Z} : xR0\} = \{\ldots, -6, -3, 0, 3, 6, \ldots\} \\ Z_1 &= \{x \in \mathbb{Z} : xR1\} = \{\ldots, -5, -2, 1, 4, 7, \ldots\} \\ Z_2 &= \{x \in \mathbb{Z} : xR2\} = \{\ldots, -4, -1, 2, 5, 8, \ldots\} \end{aligned}$$

Thus $\mathbb{Z} = Z_0 \cup Z_1 \cup Z_2$ and all are disjoint subsets.

For a given partition $S_1 \cup S_2 \cup \cdots \cup S_n$ of a set S into disjoint subsets, one can construct an equivalence relation R on S by xRy if $x, y \in S_i$ for some i.

Equivalence relation is used in almost all branches of higher mathematics.

Exercise - 1.2

1. Discuss the following relations for reflexivity, symmetricity and transitivity:
   (i) The relation R defined on the set of all positive integers by "mRn if m divides n".
   (ii) Let P denote the set of all straight lines in a plane. The relation R defined by "ℓRm if ℓ is perpendicular to m".
   (iii) Let A be the set consisting of all the members of a family. The relation R defined by "aRb if a is not a sister of b".
   (iv) Let A be the set consisting of all the female members of a family. The relation R defined by "aRb if a is not a sister of b".
   (v) On the set of natural numbers the relation R defined by "xRy if $x + 2y = 1$".
2. Let $X = \{a, b, c, d\}$ and $R = \{(a, a), (b, b), (a, c)\}$. Write down the minimum number of ordered pairs to be included to R to make it
   (i) reflexive (ii) symmetric (iii) transitive (iv) equivalence
3. Let $A = \{a, b, c\}$ and $R = \{(a, a), (b, b), (a, c)\}$. Write down the minimum number of ordered pairs to be included to R to make it
   (i) reflexive (ii) symmetric (iii) transitive (iv) equivalence
4. Let P be the set of all triangles in a plane and R be the relation defined on P as aRb if a is similar to b. Prove that R is an equivalence relation.
5. On the set of natural numbers let R be the relation defined by aRb if $2a + 3b = 30$. Write down the relation by listing all the pairs. Check whether it is
   (i) reflexive (ii) symmetric (iii) transitive (iv) equivalence
6. Prove that the relation "friendship" is not an equivalence relation on the set of all people in Chennai.
7. On the set of natural numbers let R be the relation defined by aRb if $a + b \leq 6$. Write down the relation by listing all the pairs. Check whether it is
   (i) reflexive (ii) symmetric (iii) transitive (iv) equivalence